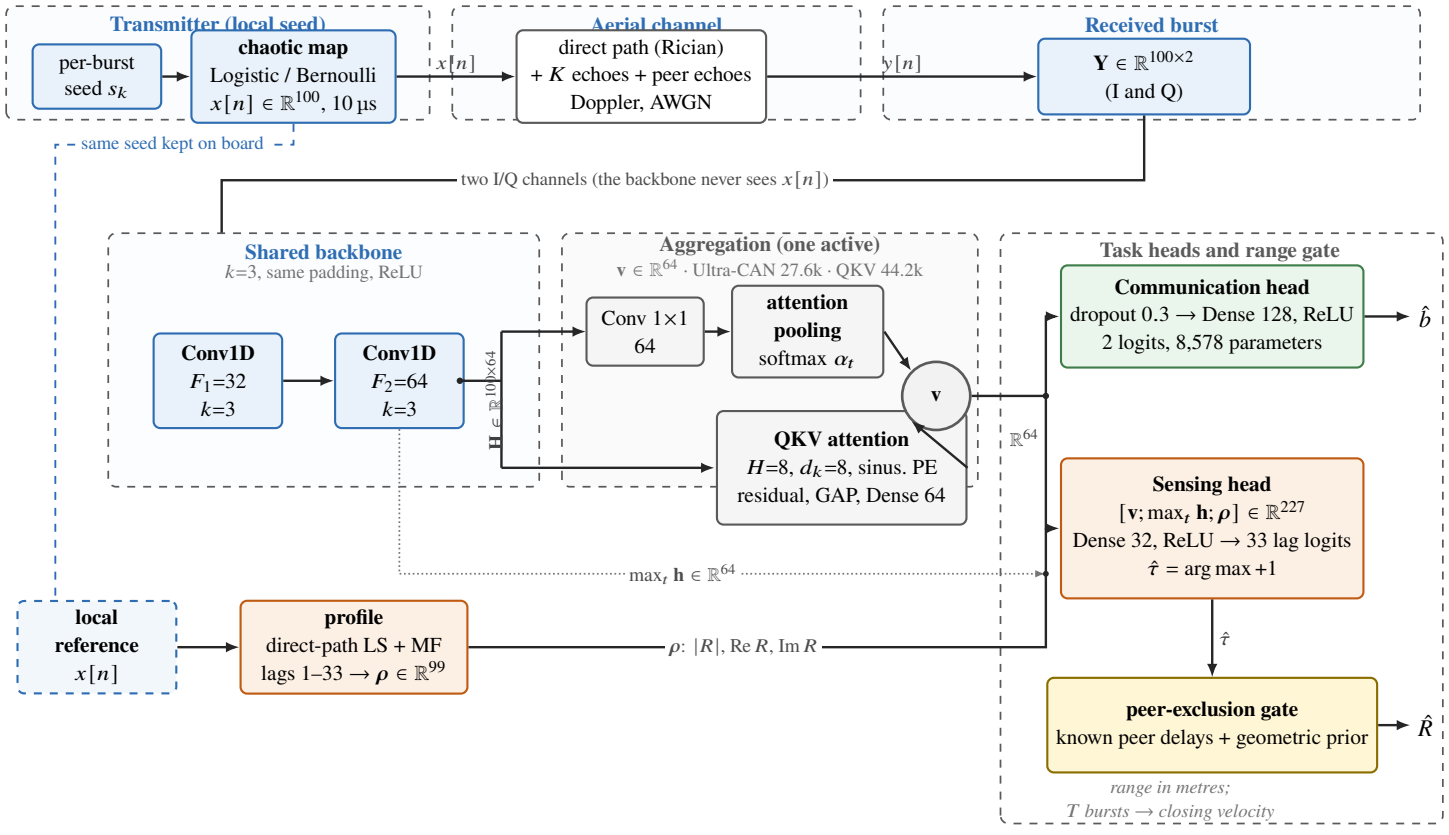

Transmitter (local seed)
per-burst seed s_k
chaotic map
Logistic / Bernoulli
$x[n] \in \mathbb{R}^{100}$, 10 µs
$x[n]$
Aerial channel
direct path (Rician)
+ K echoes + peer echoes
Doppler, AWGN
$y[n]$
Received burst
$\mathbf{Y} \in \mathbb{R}^{100\times2}$
(I and Q)
same seed kept on board
two I/Q channels (the backbone never sees $x[n]$)
Shared backbone
k=3, same padding, ReLU
Conv1D
F_1=32
k=3
Conv1D
F_2=64
k=3
$\mathbf{H} \in \mathbb{R}^{100\times64}$
Aggregation (one active)
$\mathbf{v} \in \mathbb{R}^{64}$ · Ultra-CAN 27.6k · QKV 44.2k
Conv 1×1
64
attention pooling
softmax α_t
v
QKV attention
H=8, d_k=8, sinus. PE
residual, GAP, Dense 64
$\mathbb{R}^{64}$
Task heads and range gate
Communication head
dropout 0.3 → Dense 128, ReLU
2 logits, 8,578 parameters
$\hat{b}$
Sensing head
$[\mathbf{v}; \max_t \mathbf{h}; \boldsymbol{\rho}] \in \mathbb{R}^{227}$
Dense 32, ReLU → 33 lag logits
$\hat{\tau}$ = arg max +1
$\max_t \mathbf{h} \in \mathbb{R}^{64}$
$\hat{\tau}$
local reference
$x[n]$
profile
direct-path LS + MF
lags 1–33 → $\boldsymbol{\rho} \in \mathbb{R}^{99}$
ρ: $|R|$, Re R, Im R
peer-exclusion gate
known peer delays + geometric prior
$\hat{R}$
range in metres;
T bursts → closing velocity